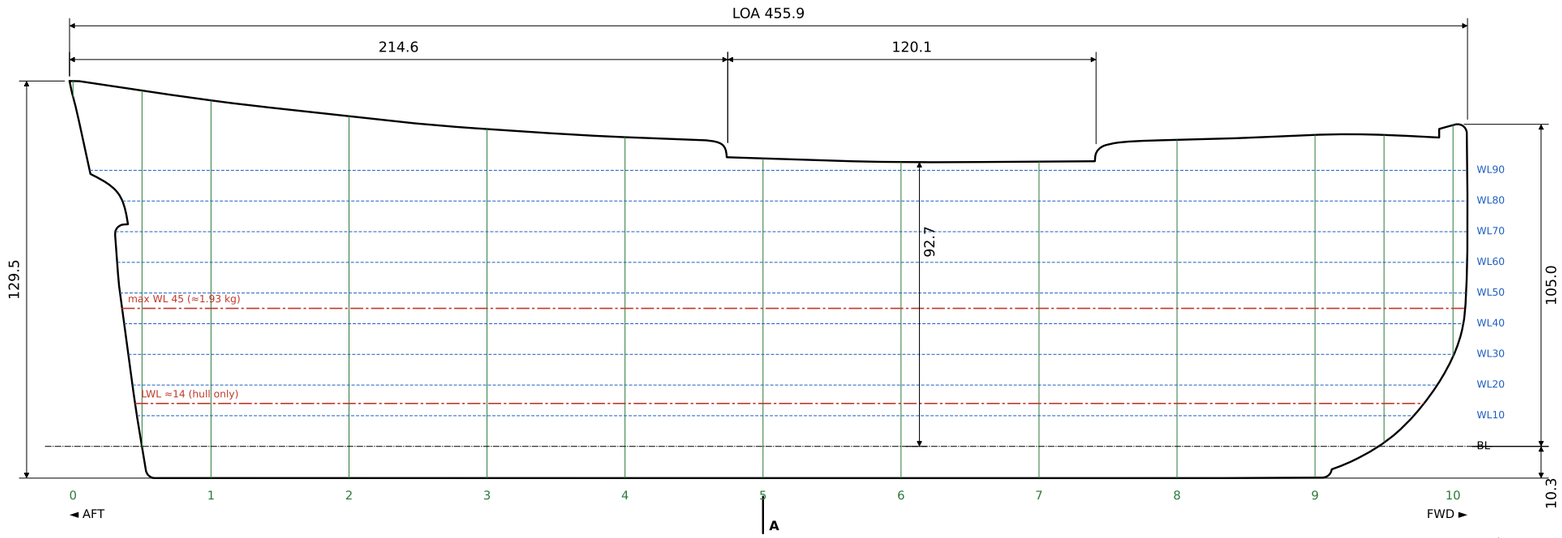
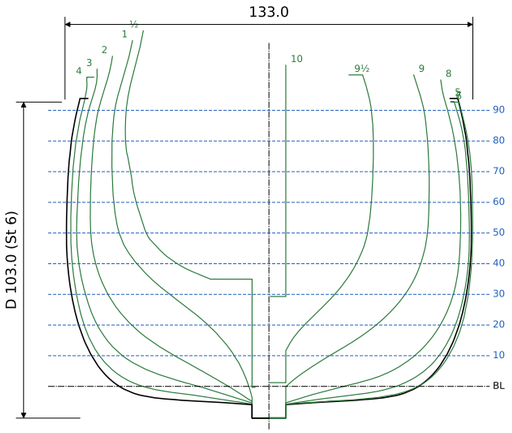
SIDE ELEVATION (sheet 1:2)

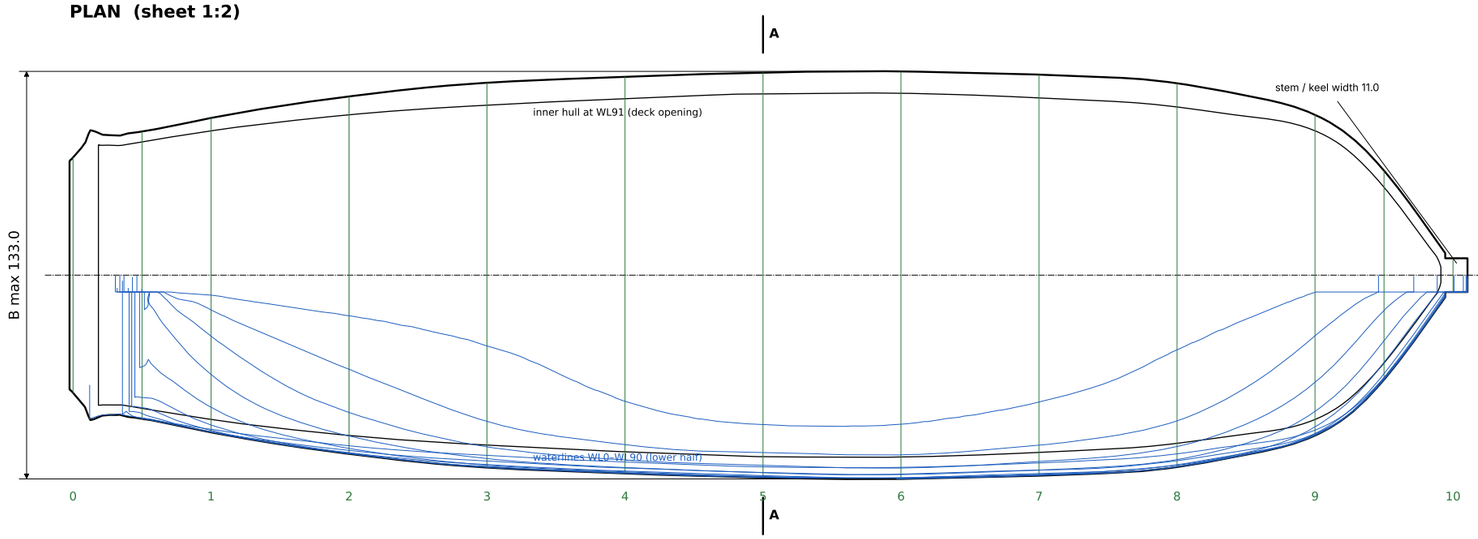


BODY PLAN (sheet 1:2)

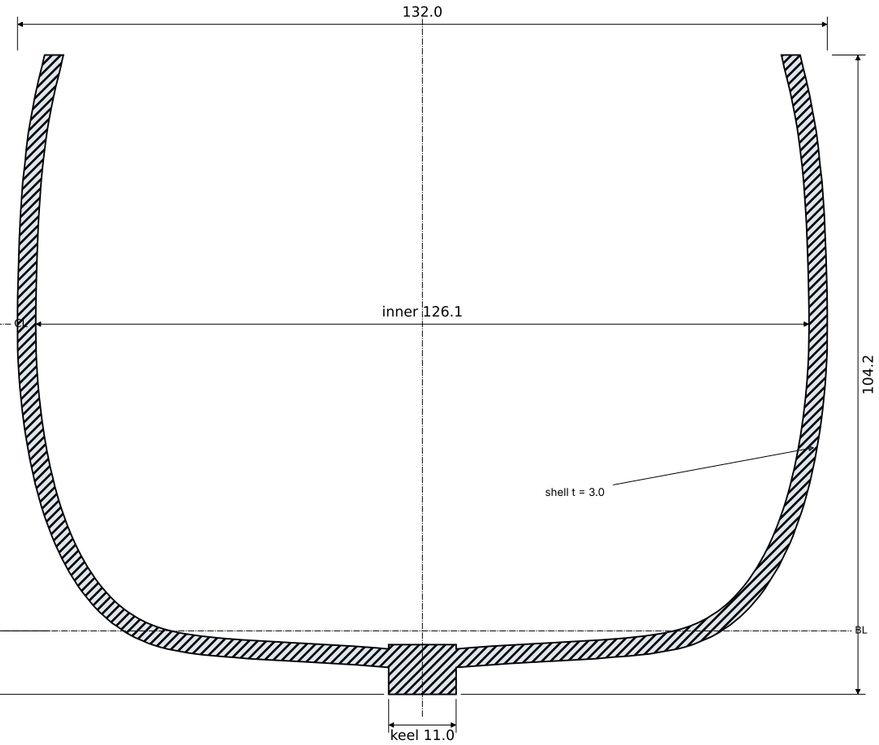
aft half St 0–5 left | fwd half St 5–10 right



PLAN (sheet 1:2)







SECTION A–A at St 5 (x = 225) (sheet 1:1)

PRINCIPAL DIMENSIONS — model scale 1 : 20

	model (mm)	full size (m)
Length overall (LOA)	455.9	9.12
Design length (inner, St 0–10)	450.0	9.00
Breadth max, outer	133.0	2.66
Breadth max, inner (design)	127.0	2.54
Depth at St 6 (keel bottom → sheer)	103.0	2.06
Height overall	129.5	2.59
Keel bottom below BL	10.3	0.21
Keel / stem / sternpost width	11.0	0.22
Station spacing	45.0	0.90
Shell thickness (model)	3.0	—

Material volume 432 cm³ · hull mass ≈ 550 g PETG / 537 g PLA

HYDROSTATICS

WL above BL	model	full	model fresh	full sea
	mm	m	g	t
	10	0.20	403	3.3
hull only	14	0.28	550	4.5
	15	0.30	586	4.8
	20	0.40	784	6.4
	25	0.50	996	8.2
	30	0.60	1218	10.0
	35	0.70	1450	11.9
	40	0.80	1689	13.8
max	45	0.90	1934	15.9

full size = model × 20³ × 1.025 (sea water)

NOTES

1. Model at scale 1:20 of a 9.0 m boat (design length 450 mm). Full-size values = model × 20 (length), × 20³ (mass). All dimensions in mm unless noted.
2. Inner skin = DELFTship design surface (hull form "ShipHai" lines, scaled to 9.0 m); 3 mm shell offset outward. Drawn from ShipHai_45cm_shell3mm.step.
3. Underwater form below BL+45 unchanged; topsides stretched so depth at St 6 = 100 mm (design keel → sheer).
4. Keep total model mass ≤ 1.9 kg (WL ≤ 45) for design accuracy. 5. Prop aperture filled by keel plate; cut later if needed.
6. Open deck; sheer steps at poop (x ≈ 214) and forecastle (x ≈ 334). Stations every 45 mm (0.9 m full size), St 0 at x = 0.

FISHING BOAT HULL — SCALE MODEL 1:20

Full size 9.0 m · model LOA 45 cm · shell 3 mm · lines & general arrangement

PROJECT	DRAWN BY
EIE Senior Project — KMUTT	Grace

SCALE	UNITS	DATE
model 1:20 · sheet 1:2	mm (model)	2026-10-10

MATERIAL	MASS (hull)	DWG No. / SHEET
PETG / PLA	≈ 550 g PETG	SH-45-001 A3 1/1 rev B